The Intersection between AI and the Climate and Biodiversity Crisis

By Constanza Gómez Mont, Founder & President, C Minds and Co-Founder, AI for Climate Global Initiative

There is no doubt that we are currently facing an urgent climate and biodiversity crisis. We need the action of all sectors and disciplines, and the best tools we have developed as humanity to accelerate change before it's too late. AI-based technologies are among the most transformative tools that exist today. The question is, how can they best be harnessed to help fight climate change and accelerate biodiversity conservation? In this module, I will discuss the intersection between the AI, conservation and climate ecosystems, showcasing interesting applications of AI in the field and unpacking important ethical considerations when it comes to using AI to address the climate and biodiversity crisis.

Lecture Transcript

0:00 Hi, I am Constanza Gómez Mont, founder and president of the action tank, C Minds, and the AI for Climate Global Initiative. Today we're going to talk about the intersection between AI and the climate and biodiversity crisis. Discussing both the profound opportunities AI brings to advancing the solution to these, and at the same time, the equally important risks the use of AI presents. As we know, climate change affects everything from environmental wellness, to economies, geopolitics, migration, health and food availability among a long list. And sadly, as discussed in COP26, the world is failing to limit global warming to 1.5 degrees Celsius above pre industrial levels, despite promises to do so. Climate change affects every region and all communities, especially people in vulnerable situations. Moreover, apart from the climate crisis, we have a profound biodiversity crisis that is destabilizing Earth's systems. There is no doubt that we're dealing with an urgent challenge that needs the action of all sectors and disciplines. And we need the best tools that we have developed as humanity to be able to accelerate the needed change. AI based technologies are one of the most transformative tools that exist today. And the question is, how can they best be harnessed to help fight climate change and accelerate biodiversity conservation? For example, machine learning based technologies can be powerful tools in reducing greenhouse gas emissions, manage natural disasters in a smarter and more effective way, enable a more efficient use of energy, help discover greener materials for construction, and help monitor species among various other types of applications. There's much to learn in this intersection and time is ticking. This is why in C Minds, the action tank I lead, we founded the AI for Climate Global Initiative aimed at advancing interdisciplinary conversations between the AI, conservation and climate ecosystems, showcasing interesting applications and helping pilot initiatives. Let's explore some of these use cases we have mapped out to give an example of what is out there and the future potential. Let's start with agriculture. Agriculture has been one of the sectors that has been affected greatly by the changing climate. Rising temperatures and changing precipitation patterns are increasingly putting crop yields and food availability and accessibility at risk. To give one example, two degrees Celsius of local warming will lead to production losses for wheat, rice and corn across the world, negatively disrupting food security. Also, food production will have to increase by 60% to feed an additional 2 billion people by 2050. And this means that we need greater effectiveness in the entire food supply chain. AI is helping solve some of these pressing challenges. For example, sensors and other data gathering tools are allowing farms to produce and gather 1000s of data points on weather conditions, temperature, water consumption, soil, among others. With the help of AI systems, this vast amount of data is being harnessed in real time to obtain insights that allow for better management, risk detections and better efficiencies. Did you know that choosing the right period or even exact days to plant seeds is a decision that makes or breaks the profitability of that year's harvest. Farmers are benefiting from AI based tools that give key information and precise dates for sowing the seeds for maximum yields. Moreover, there are more and more applications of machine learning tools that are giving insights on soil health, fertilizer recommendations, information on how the weather conditions will affect the crops and price forecasts that help farmers obtain maximum profit. Other applications in play helping detect exact areas that have been infected with weeds and pests, reducing usage of herbicides and managing harvest loss in a more timely way. Robots that use computer vision and AI are allowing, for example, eliminating 80% of the volume of chemicals that are normally sprayed on the crops and bringing down expenditure of herbicide by 90% in some cases.

4:39 And these models are not only being used by large agricultural companies, but by small farmers around the world. For example, the International Crops Research Institute for the Semi Arid Tropics, built a machine learning model downloaded by 12 million users globally that allows farmers to diagnose pest damage, plant disease and nutrient deficiencies by taking a photo of their affected crop. Detection success is now at 80% accuracy they say. FAO estimates that between 20 and 40% of global crop yields are reduced each year because of plant diseases and pests, and thus, these types of AI applications become key assets. On other topics we're seeing impactful AI based case studies that address biodiversity conservation in various latitudes of the world. For example, AI software is helping rangers predict poaching hotspots in wildlife parks. There is a case study that we have loved to showcase in our AI for Climate Lightning Talks, called Protection Assistant for Wildlife Security - PAWS in short. They developed in one of Harvard's labs, in collaboration with World Wildlife Fund, Wildlife Conservation Society and other organizations that are part of Smart Partnership. PAWS takes historical data of the rangers and correlates it to geographical information about the park: how far certain areas are from roads and rivers; the slope of the land; precipitation level; distance to international borders; wildlife counts; and then builds predictive models based on past poaching activity in squares of one kilometer by one kilometer. In Cambodia, for example, the use of these systems allows rangers to find and remove poachers five times more than previously done. Other conservation challenges include illegal deforestation. Actually, the UN states that up to 90% of the logging of tropical rainforest is illegal. And thus addressing this becomes an opportunity for innovation. To this end, sound recognition models that identify the sound of saw chains and give real time alerts are being harnessed by organizations such as the Rainforest Connection. And they are also using these acoustic monitoring machine learning systems to help monitor species such as the spider monkey in Latin America. For this work, they are partnered up with Huawei to use recycled phones to capture sounds, and through sound processing systems, better understand the monkeys behavior and movements to have more information to understand them and protect them. And all that data is being used to create a digital library to give global scientists instant access to vast amounts of raw, acoustic data of rainforest so they can accelerate the creation of specific applications for species monitoring and protection. And satellite data is also being harnessed to manage the conservation of wildlife more effectively as well. For example, the organization, Restore, uses satellite images and other data sources to give people actionable information for precision agroforestry and other reforestation activities. It provides high resolution satellite imagery down to 50 centimetres detailed enough to show individual trees. This means change over time, can be monitored remotely in almost any terrestrial location. They have now more than 73,000 registered restoration projects and work with a scientific group of more than 200 members. So there's a huge potential as well in the use of AI to promote a more transparent and accountable carbon credit sector, a market that could be worth more than 50 billion US Dollars in 2030. For example, the startup Victoria will be harnessing AI systems to better measure the carbon capture of natural reserves, which in the past has been a labor intensive and manual process, enabling landowners and investors to have certainty of the impact of their credits. It will also harness these technologies to help pioneer, together with other organizations, a biodiversity conservation market, based in real time data backed impact metrics. My co-founder Christopher has a motto which I love - "Let's make visible the biodiversity that sustains us". And there is no better way to do this than to harness AI. These are only very few examples of the growing list of applications of AI systems to address the climate and biodiversity crisis. Actually, a 2020 UN study established that AI could positively enable 93% of the environmental targets. Innovation is a must in this field.

9:37 However, AI's negative impact and ethical considerations are equally important to understand and address. One of the most important ones is its energy consumption. All the advancements of AI require vast amounts of computing power and electricity to devise and train algorithms. And there are growing concerns that if this is not addressed, these technologies will only deepen the climate crisis. Training powerful machine learning algorithms sometimes represents running large banks of computers for days or even weeks. And fine tuning is also computationally intensive. Researchers at the University of Massachusetts, Amherst recently conducted a study assessing the energy consumption required to train several common large AI models. The study revealed that the training process can emit nearly five times more the lifetime emissions of an average car or the equivalent of about 300 round trip flights between New York and San Francisco. Moreover, some researchers are predicting that by 2030, computing and communications technology will consume between 8% and 20% of the world's electricity, with data centers accounting for a third of that. For example, GPT-3, our recent powerful language model by Open AI is estimated to have consumed enough energy in training to have a carbon footprint that equals to driving a car from Earth to the moon and back. As you can see, the impact is no small thing. Large big Tech's have actually pledged to be AI carbon neutral in some years. However, we need to mainstream this effort to include all AI based companies. Needed actions include data centers to base your operations on renewable energy, and cloud providers to be transparent on the energy consumption of machine learning systems. Also, we need to work towards an evolution of risk evaluation models to include a multi dimensional framework that includes environmental impact. And tools to evaluate the environmental impact, such as the one Mila created, are being developed to help with this. However, the task of carbon impact measurement is not easy. As future energy consumption of more complex and dynamic models are not easy to predict, coupled with the fact that the measurement of carbon footprint of AI is multi dimensional, including the physical data center infrastructure, and where the energy is sourced from. However, we as an AI ecosystem have to figure this out. For AI risk evaluation frameworks need to include environmental impact assessments. Another challenge that has to be addressed is the extraction of raw materials that is needed to sustain the growth of the digital sector. The extraction of nickel, cobalt, graphite for lithium ion batteries used for example, in electrical cars powered by AI, and smartphones has already damaged the environment and will increase to damage natural sites if it is not correctly managed. We need to prevent the unsustainable exploitation, use and transformation of natural resources contributing to the deterioration of the environment and the degradation of ecosystems. As experts say, sustainability must be treated as a key success indicator in any AI based project. At the end, we need AI to be smarter, but greener and more transparent about their environmental impact.

13:16 The third challenge I want to mention is accessibility. When we're talking about AI applications for better farming, better research management, better biodiversity conservation, among others, we're normally talking about the work of communities in areas where there is little connectivity and thus access to sophisticated tools becomes a challenge. AI tools have to be inclusive for local communities that can benefit from them most. And this implies not only digital connectivity and access but also digital literacy. And lastly, inclusion of local communities, especially indigenous communities, have to be part of the entire lifecycle of AI. Indigenous people inhabit nearly 20% of the planet, mainly in areas where they have lived for 1000s of years. As WWF says indigenous peoples are among the Earth's most important stewards, as evidenced by the high degree of correspondence between the land, waters and territories of indigenous peoples and the remaining high biodiversity regions of the world. In any scenario where AI is being applied to address environmental concerns in these lands, or their resource management, it is imperative that indigenous peoples that inhabit these territories have a seat at the table. We cannot talk about an ethical use of AI for climate and conservation if it is not inclusive at heart. The risks and challenges are as big as the opportunities of AI in this field. And even though we have a long way to go, they're important global efforts that are paving the way for an environmentally responsible use of AI. For example, in 2020, I had the privilege of being part of a group of 24 experts from all corners of the world to develop, through UNESCO lead and the input of a global consultation, the first global instrument of AI ethics, which was the first document of the type to address AI's role in the environment. It recognizes that AI technologies have the potential to be beneficial to the environment and ecosystems. And in order for those benefits to be realized, potential harms, and negative impacts on the environment and ecosystems should not be ignored, but instead addressed. The instrument was formally adopted by UNESCO's General Conference in November 2021. There are also other great initiatives such as the Center for AI and Climate. The efforts done for the AI climate roadmap for the Global Partnership on AI, and Climate Change AI that showcases a vast number of ML possible use cases, among other initiatives. On the overall inclusion component of AI, key special like frameworks are being developed, such as the one from the World Economic Forum Global Future Council on AI for Humanity, that will be published in the summer of 2023, among other resources developed by key institutions in all regions of the world that can be found in the AI Fairness Global Library and other repositories. As for Latin America, the region where I am from, we have partnered up with the Inter-American Development Bank to develop, under their initiative, fAIr LAC, various tools in Spanish and Portuguese for both governments and intrapreneurs that help address AI ethical considerations that can help actors in the AI for climate and biodiversity field as well. And last but not least, the AI for Climate Initiative, founded by C Minds and the Christopher Córdova Agency, helps bridge the worlds of AI, climate and biodiversity conservation with a human rights perspective. You can check out more information of the use cases presented today and many others in the AI for Climate website. There is no doubt that AI has a profound potential to accelerate solutions for the climate and biodiversity crisis. However, its role as a contributor to climate change cannot be overlooked. In this sense, an ethical and human based rights approach to AI necessarily needs to include an environmental perspective to ensure that AI is sustainable and benefits our ecosystems as a whole. *¡Muchas gracias!*